Tutorial Sheet 5: Linear Maps I

Getting started.

1. (a) Yes, this map is linear. Indeed, we have for $\vec{v} \in \mathbb{R}^2$,

$$T_1(\vec{v}) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \vec{v},$$

hence T_1 is the map associated to the matrix $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and we know from the course that such maps are linear.

(b) No: For example, under scaling $T_2(2\begin{pmatrix} 1 \\ 1 \end{pmatrix}) = \begin{pmatrix} 4 \\ 4 \end{pmatrix} \neq 2T_2(\begin{pmatrix} 1 \\ 1 \end{pmatrix}) = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$.

(c) No: For example, under scaling $T_3(2x) = 4x^2 \neq 2T_3(x) = 2x^2$.

(d) Yes, this map is linear. First we need to show linearity under addition. Let $P(x), Q(x) \in \mathcal{P}(\mathbb{R})$ and let $\lambda \in \mathbb{R}$. We have, by linearity of the integral:
$T_4(P+Q) = \int_0^1 (P(t)+Q(t))e^t \mathrm{d}t = \int_0^1 (P(t)e^t + Q(t)e^t)\mathrm{d}t = \int_0^1 P(t)e^t \mathrm{d}t + \int_0^1 Q(t)e^t \mathrm{d}t = T_4(P) + T_4(Q)$.
We also have, by linearity of the integral:
$T_4(\lambda P) = \int_0^1 (\lambda P(t))e^t \mathrm{d}t = \int_0^1 \lambda P(t)e^t \mathrm{d}t = \lambda \int_0^1 P(t)e^t \mathrm{d}t = \lambda T_4(P)$.

2. (a) We can see that $T : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x-y \\ 2x+y \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$, and we can read off the matrix of T from there.
Another method is to calculate the images of the vectors in the given basis:
$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 1 \end{pmatrix}$
Therefore the matrix encoding this transformation is
$\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$.

(b) We compute the image of the basis vectors $1, x, x^2, x^3$:
$1 \mapsto x^3$
$x \mapsto -x + 2x^3$
$x^2 \mapsto -2x^2 + 4x^3$
$x^3 \mapsto 5x^3$
Therefore the matrix of T in that basis is

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 1 & 2 & 4 & 5 \end{pmatrix}$$

(c) We compute the image of the basis vectors $1, x, \ldots, x^n$:

$1 \mapsto 0$

$x \mapsto 1$

$x^2 \mapsto 2x$

$\vdots$

$x^n \mapsto nx^{n-1}$

Therefore the matrix of T in that basis is

$$\begin{pmatrix} 0 & 1 & 0 & \ldots & \ldots & 0 \\ 0 & 0 & 2 & \ldots & \ldots & 0 \\ 0 & 0 & 0 & \ddots & \ddots & 0 \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ \vdots & \vdots & \vdots & & & n \\ 0 & 0 & 0 & \ldots & \ldots & 0 \end{pmatrix}$$

3. (a) We apply the algorithm to compute the inverse of a matrix:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 3 & -1 & -2 & 0 & 1 & 0 \\ 2 & -9 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -7 & -5 & -3 & 1 & 0 \\ 0 & -13 & -2 & -2 & 0 & 1 \end{array}\right)$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -7 & -5 & -3 & 1 & 0 \\ 0 & 0 & 51 & 25 & -13 & 7 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -7 & -5 & -3 & 1 & 0 \\ 0 & 0 & 1 & \frac{25}{51} & -\frac{13}{51} & \frac{7}{51} \end{array}\right)$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{4}{51} & \frac{2}{51} & -\frac{5}{51} \\ 0 & 0 & 1 & \frac{25}{51} & -\frac{13}{51} & \frac{7}{51} \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{6}{17} & \frac{3}{17} & \frac{1}{17} \\ 0 & 1 & 0 & \frac{4}{51} & \frac{2}{51} & -\frac{5}{51} \\ 0 & 0 & 1 & \frac{25}{51} & -\frac{13}{51} & \frac{7}{51} \end{array}\right)$$

So A is invertible and $A^{-1} = \frac{1}{51}\begin{pmatrix} 18 & 9 & 3 \\ 4 & 2 & -5 \\ 25 & -13 & 7 \end{pmatrix}$

(b) We apply the algorithm to compute the inverse of a matrix:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & -6 & -2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right)$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -1 & -6 & -2 & 1 & 0 \\ 0 & 0 & -4 & -2 & 1 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{2} & \frac{-1}{4} & \frac{-9}{4} \\ 0 & 1 & 0 & -1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{array}\right)$$

so B is invertible and $B^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{-1}{4} & \frac{-9}{4} \\ -1 & \frac{1}{2} & \frac{3}{2} \\ \frac{1}{2} & -\frac{1}{4} & -\frac{1}{4} \end{pmatrix}$

4. We apply the algorithm to compute the inverse of a matrix:

$$\left(\begin{array}{cccc|cccc} 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -2 & -1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 & 0 & -1 & 0 & 1 \end{array}\right)$$

$$\to \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & -2 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & -2 & 0 & -1 & 0 & 1 \end{array}\right) \to \left(\begin{array}{cccc|cccc} 1 & -1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 0 & -2 & 0 & -1 & 0 & 1 \end{array}\right)$$

$$\to \left(\begin{array}{cccc|cccc} 2 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & 0 & -1 & 0 & 1 & -1 \\ 0 & 0 & 0 & -2 & 0 & -1 & 0 & 1 \end{array}\right) \to \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/2 & 0 & 1/2 & 1/2 \\ 0 & 1 & 0 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & -1/2 & 1/2 \\ 0 & 0 & 0 & 1 & 0 & 1/2 & 0 & -1/2 \end{array}\right)$$

Therefore A is invertible and $A^{-1} = \begin{pmatrix} 1/2 & 0 & 1/2 & 1/2 \\ 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & -1/2 & 1/2 \\ 0 & 1/2 & 0 & -1/2 \end{pmatrix}$

We have $A\vec{x} = \vec{y} \Rightarrow \vec{x} = A^{-1}\vec{y}$, so we get

$$\text{(a) } \vec{x} = \begin{pmatrix} 3/2 \\ 1 \\ 1/2 \\ 0 \end{pmatrix} \text{ (b) } \vec{x} = \begin{pmatrix} 4 \\ 3 \\ 1 \\ -1 \end{pmatrix}.$$

5. **Polynomial equations.**

   (a) For $P, Q \in \mathcal{P}_2(\mathbb{R})$ and $\lambda \in \mathbb{R}$, we have:
   $T(P+Q) = \frac{1}{2}((P+Q)(x) + (P+Q)(x+1)) = \frac{1}{2}(P(x) + P(x+1)) + \frac{1}{2}(Q(x) + Q(x+1)) = T(P) + T(Q)$.
   $T(\lambda P) = \frac{1}{2}((\lambda P)(x) + (\lambda P)(x+1)) = \lambda \times \frac{1}{2}(P(x) + P(x+1)) + = \lambda T(P)$.
   Thus, T is linear. We compute the image of the vectors of the basis $\mathcal{B} := (1, x, x^2)$:
   $T(1) = 1, T(x) = x + \frac{1}{2}, T(x^2) = x^2 + x + \frac{1}{2}$.
   Thus, the matrix of T in the basis $\mathcal{B} := (1, x, x^2)$ is

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 1/2 & 1/2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

   (b) $[T]_{\mathcal{B}}$ is a square matrix in row echelon form with non-zero diagonal coefficients, so it is invertible. We use the standard algorithm to compute its inverse, which yields

$$\left(\begin{array}{ccc|ccc} 1 & 1/2 & 1/2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1 & 0 & -1/2 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1/2 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

so

$$[T]_{\mathcal{B}}^{-1} = \begin{pmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

(c) Solving $\frac{1}{2}(P(x) + P(x+1)) = x^2$ can be rephrased as solving $T(P) = x^2$, whose solution is $P(x) = T^{-1}(x^2)$. We use the matrix representation to perform the computations. We have

$$[x^2]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightsquigarrow [P(x)]_{\mathcal{B}} = [T]_{\mathcal{B}}^{-1}[x^2]_{\mathcal{B}} = \begin{pmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix},$$

so $P(x) = x^2 - x$.

Similarly, to solve $T(Q) = x^2 + x + 1$, we compute

$$[x^2 + x + 1]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \rightsquigarrow \quad [Q(x)]_{\mathcal{B}} = \begin{pmatrix} 1 & -1/2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix},$$

so $Q(x) = x^2 + 1/2$.

6. **Solutions of a homogeneous linear differential equation.** We consider the set W of functions $y \in \mathcal{C}^\infty(\mathbb{R})$ such that

$$y^{(3)} + y^{(2)} + y' + y = 0.$$

(a) The zero function is clearly in W. Let $f,\ g \in W$ and $\lambda \in \mathbb{R}$. Consider

$$(f+g)^{(3)}+(f+g)^{(2)}+(f+g)'+f+g = f^{(3)}+f^{(2)}+f'+f+g^{(3)}+g^{(2)}+g'+g = 0+0 = 0.$$

Similarly $(\lambda f)^{(3)} + (\lambda f)^{(2)} + (\lambda f)' + \lambda f = 0$.

Thus, W is a subspace of $\mathcal{C}^\infty(\mathbb{R})$.

(b) First, $T(f+g) = \big(f(0)+g(0), f'(0)+g'(0), f^{(2)}(0)+g^{(2)}(0)\big)^T = T(f) + T(g)$

Similarly $T(\lambda f) = \big(\lambda f(0), \lambda f'(0), \lambda f^{(2)}(0)\big)^T = \lambda T(f)$, so T is a linear map.

(c) For $y(x) = \cos(x)$, we have

$$y^{(3)}(x) + y^{(2)}(x) + y'(x) + y(x) = \sin(x) - \cos(x) - \sin(x) + \cos(x) = 0,$$

so $\cos(x) \in W$.

For $y(x) = \sin(x)$, we have

$$y^{(3)}(x) + y^{(2)}(x) + y'(x) + y(x) = -\cos(x) - \sin(x) + \cos(x) + \sin(x) = 0,$$

so $\sin(x) \in W$.

For $y(x) = e^{-x}$, we have

$$y^{(3)}(x) + y^{(2)}(x) + y'(x) + y(x) = -e^{-x} + e^{-x} - e^{-x} + e^{-x} = 0,$$

so $e^{-x} \in W$.

We now compute

$$T(\cos(x)) = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \ , \ T(\sin(x)) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \ , \ T(e^{-x}) = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

To test whether this forms a basis of $\mathbb{R}^3$, we compute the determinant of the associated matrix, and we find:

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix} = 2 \neq 0.$$

It thus follows that these vectors form a basis of $\mathbb{R}^3$ by a result from the course.

(d) We are looking for a function of the form

$$f(x) = a\cos(x) + b\sin(x) + c\ e^{-x} \ ,$$

such that

$$f(0) = 3, \quad f'(0) = -2, \quad f^{(2)}(0) = -1.$$

This gives

$$a + c = 3, \quad b - c = -2, \quad -a + c = -1,$$

which we solve as usual by Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & -2 \\ -1 & 0 & 1 & -1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 2 & 2 \end{array}\right)$$

This gives $c = 1$, $b = -1$, $a = 2$ and therefore $f(x) = 2\cos(x) - \sin(x) + e^{-x}$.

7. (a) There are three types of row operations to consider:

• The row operation $R_i \leftrightarrow R_j$ amounts to replacing B with the product EB, where E is the following

$$E = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 0 & & 1 & & \\ & & & \ddots & & & \\ & & 1 & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix},$$

that is, we have $e_{kk} = 1$ for $k \neq i$ and $k \neq j$, $e_{ii} = a_{jj} = 0$, $e_{ij} = e_{ji} = 1$, and all the other coefficients are zero. This matrix is indeed invertible as its determinant is -1.

• The row operation $R_i \leftrightarrow \alpha R_i$, with $\alpha \neq 0$, amounts to replacing B with the product EB, where E is the following matrix:

$$E = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & \alpha & & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix},$$

that is $e_{kk} = 1$ for $k \neq i$, $e_{ii} = \alpha$, and all the other coefficients are zero. This matrix is indeed invertible since $\det(E) = \alpha \neq 0$.

• The row operation $R_i \leftrightarrow R_i + \alpha R_j$ amounts to replacing B with the product EB, where E is the following matrix:

$$E = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & \alpha & & \\ & & & \ddots & & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix},$$

that is, $e_{kk} = 1$ for all $1 \leq k \leq n$, $e_{ij} = \alpha$, and all the other coefficients are zero. This matrix is indeed invertible as $\det E = 1$.

Let $R_1, \ldots, R_m$ be the sequence of row operations to go from the matrix B to a matrix C in row echelon from, and let $E_1, \ldots, E_m$ be the associated invertible $n \times n$ matrices (by (c)). We thus have $B' = AB$, with $A := E_m \times \cdots \times E_1$, and A is invertible as a product of invertible matrices.

(b) Since C contains exactly ℓ pivots and is in row echelon form, the last $n - \ell$ rows of C consist of zeroes. In particular, since the vectors $A\vec{v}_1, \ldots, A\vec{v}_k$ are exactly the columns of C, it follows that $A\vec{v}_1, \ldots, A\vec{v}_k$ are contained in V. Let $\vec{x} \in W$. By definition of a span, we can write $\vec{x}$ as a linear combination of the form

$$\vec{x} = \lambda_1 \vec{v}_1 + \cdots + \lambda_k \vec{v}_k$$

for some $\lambda_1, \ldots, \lambda_k \in \mathbb{R}$. By linearity of A, we thus get

$$A\vec{x} = \lambda_1 A\vec{v}_1 + \cdots + \lambda_k A\vec{v}_k$$

and since V is a subspace of $\mathbb{R}^n$, hence stable under linear combinations, we get $A\vec{x} \in V$.

(c) Let us show first that $A\vec{u}_1, \dots, A\vec{u}_\ell$ is linearly independent. The matrix associated to the equation (in unknowns $x_1, \dots, x_\ell$)

$$x_1 A\vec{u}_1 + \cdots + x_\ell A\vec{u}_\ell = \vec{0}$$

is of the form

$$\begin{pmatrix} \star & & & & \\ 0 & \star & & & \\ 0 & 0 & \ddots & & \\ \vdots & \vdots & \ddots & \ddots & \\ 0 & 0 & \cdots & 0 & \star \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix},$$

where the entries represented by a star are non-zero by construction of $\vec{u}_1, \dots, \vec{u}_\ell$, and the coefficients above the diagonal are arbitrary are are not indicated. (This matrix is exactly the matrix obtained by removing from C the columns that do not contain a pivot.) In particular, there is no free variable, hence the family $A\vec{u}_1, \dots, A\vec{u}_\ell$ is linearly independent. We thus have a linearly independent family of ℓ vectors inside the subspace V. But the dimension of V is exactly ℓ, so by a result of the course $A\vec{u}_1, \dots, A\vec{u}_\ell$ is a basis of V.

(d) Let us first show that $\vec{u}_1, \dots, \vec{u}_\ell$ is linearly independent. Let $\lambda_1, \dots, \lambda_\ell \in \mathbb{R}$ such that

$$\lambda_1 \vec{u}_1 + \cdots \lambda_\ell \vec{u}_\ell = \vec{0}.$$

By linearity of A, we get

$$\lambda_1 A\vec{u}_1 + \cdots \lambda_\ell A\vec{u}_\ell = A\vec{0} = \vec{0}.$$

Since $A\vec{u}_1, \dots, A\vec{u}_\ell$ is linearly independent by (c), this implies that $\lambda_1 = \cdots = \lambda_\ell = 0$, hence $\vec{u}_1, \dots, \vec{u}_\ell$ is linearly independent.

Let us now show that $\vec{u}_1, \dots, \vec{u}_\ell$ spans W. Let $\vec{x} \in W$. By (b), $A\vec{x} \in V$, and since $A\vec{u}_1, \dots, A\vec{u}_\ell$ is a basis of V, there exists $\lambda_1, \dots, \lambda_\ell \in \mathbb{R}$ such that

$$A\vec{x} = \lambda_1 A\vec{u}_1 + \cdots A\lambda_\ell \vec{u}_\ell.$$

By linearity of A, we thus get

$$A\big(\vec{x} - (\lambda_1 \vec{u}_1 + \cdots \lambda_\ell \vec{u}_\ell)\big) = \vec{0}.$$

But since A is invertible, this implies that $\vec{x} - (\lambda_1 \vec{u}_1 + \cdots \lambda_\ell \vec{u}_\ell) = \vec{0}$, hence

$$\vec{x} = \lambda_1 \vec{u}_1 + \cdots \lambda_\ell \vec{u}_\ell,$$

which is what we wanted.

Thus, $\vec{u}_1, \dots, \vec{u}_\ell$ is a basis of W.